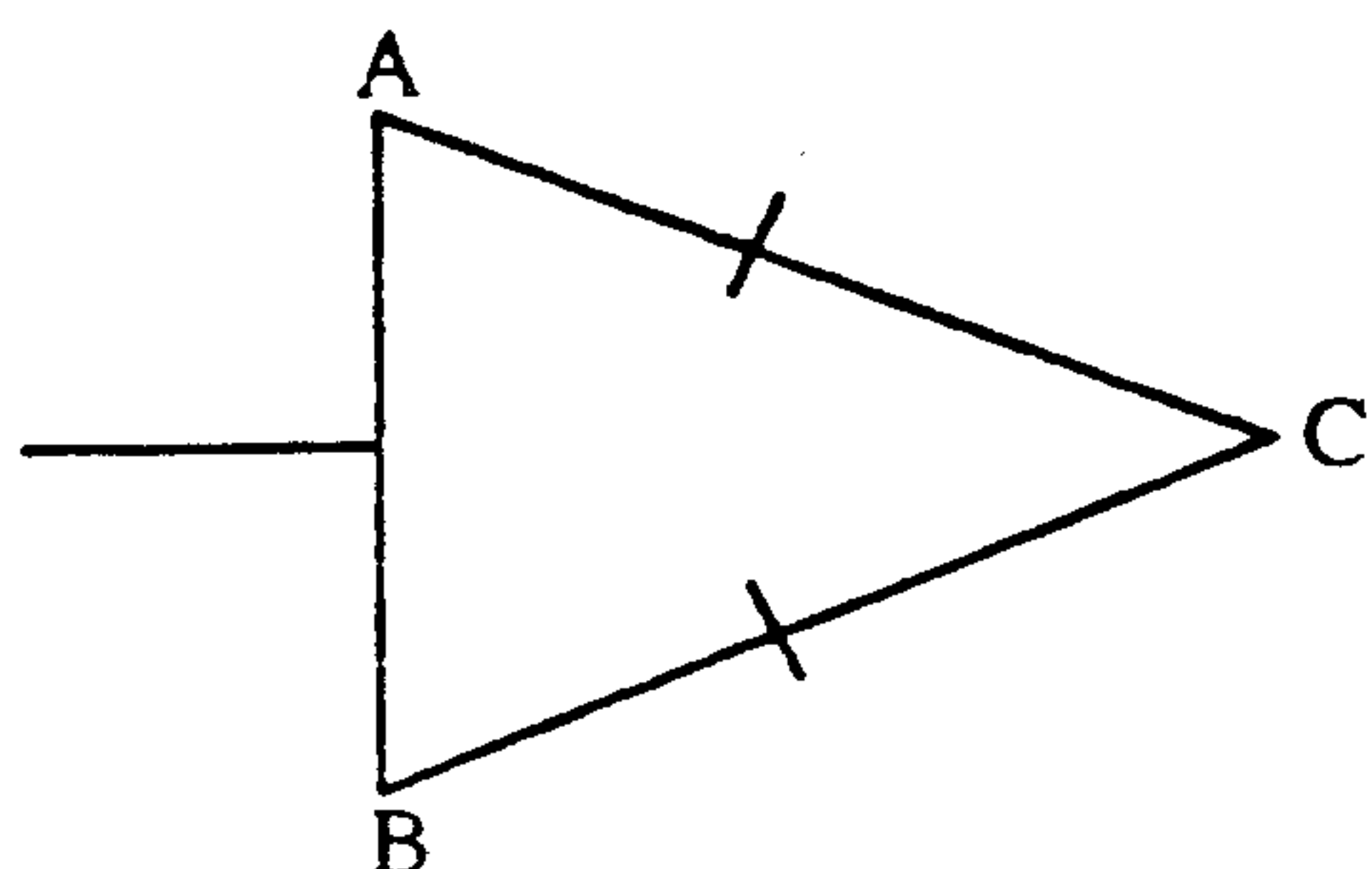


Question 1. (Start a new page.)

- a) Given that P, Q, R are the points (1,3), (2,0), and (5,1) respectively :
- show that the line PQ is perpendicular to QR;
 - prove that the triangle PQR is isosceles;
 - find the coordinates of S, such that PQRS is a square.

- b) A triangle LMN is right angled at L.
If $LM = x$, $NM = x + 3$, and $NL = 5$, find x .

c)



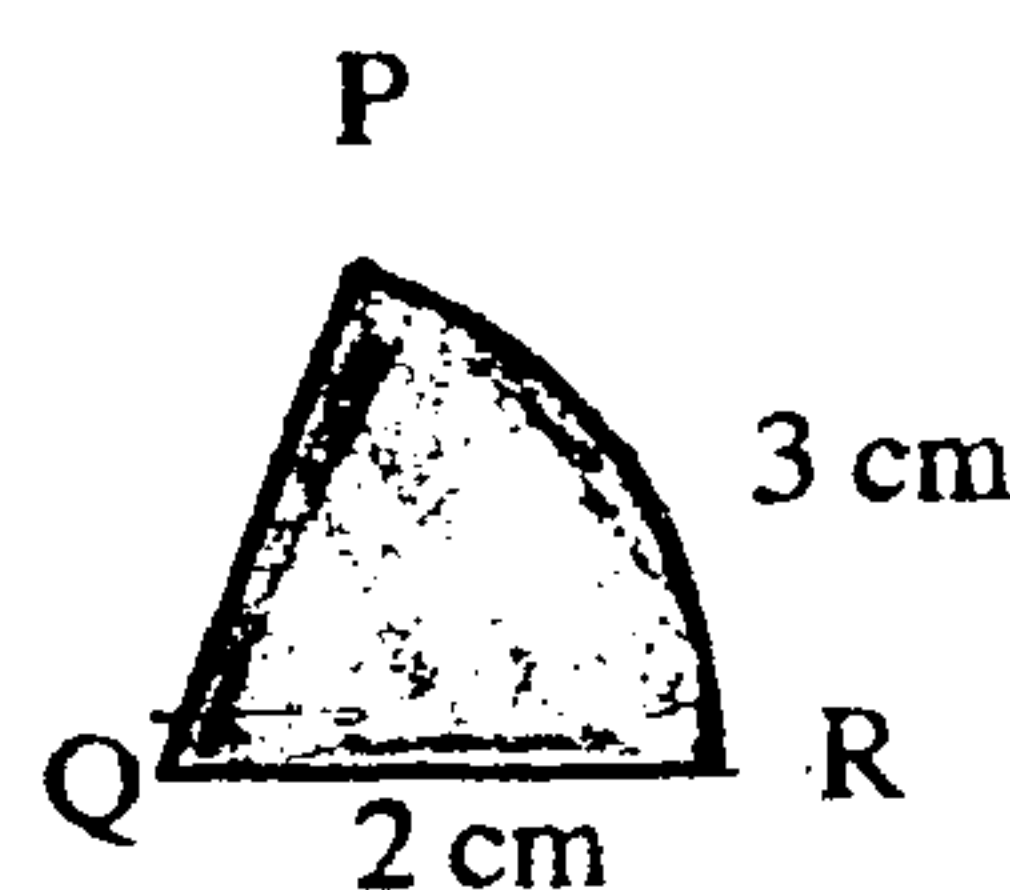
Find the exact area of the arrow head, where $AC = BC = 6$ cm and $\angle ACB = 45^\circ$.

- d) Joan invested \$5,000 at the beginning of 1985 at 8% per annum, compounded at the end of each year.
What will her investment be worth at the end of 1995 ?

Question 2. (Start a new page.)

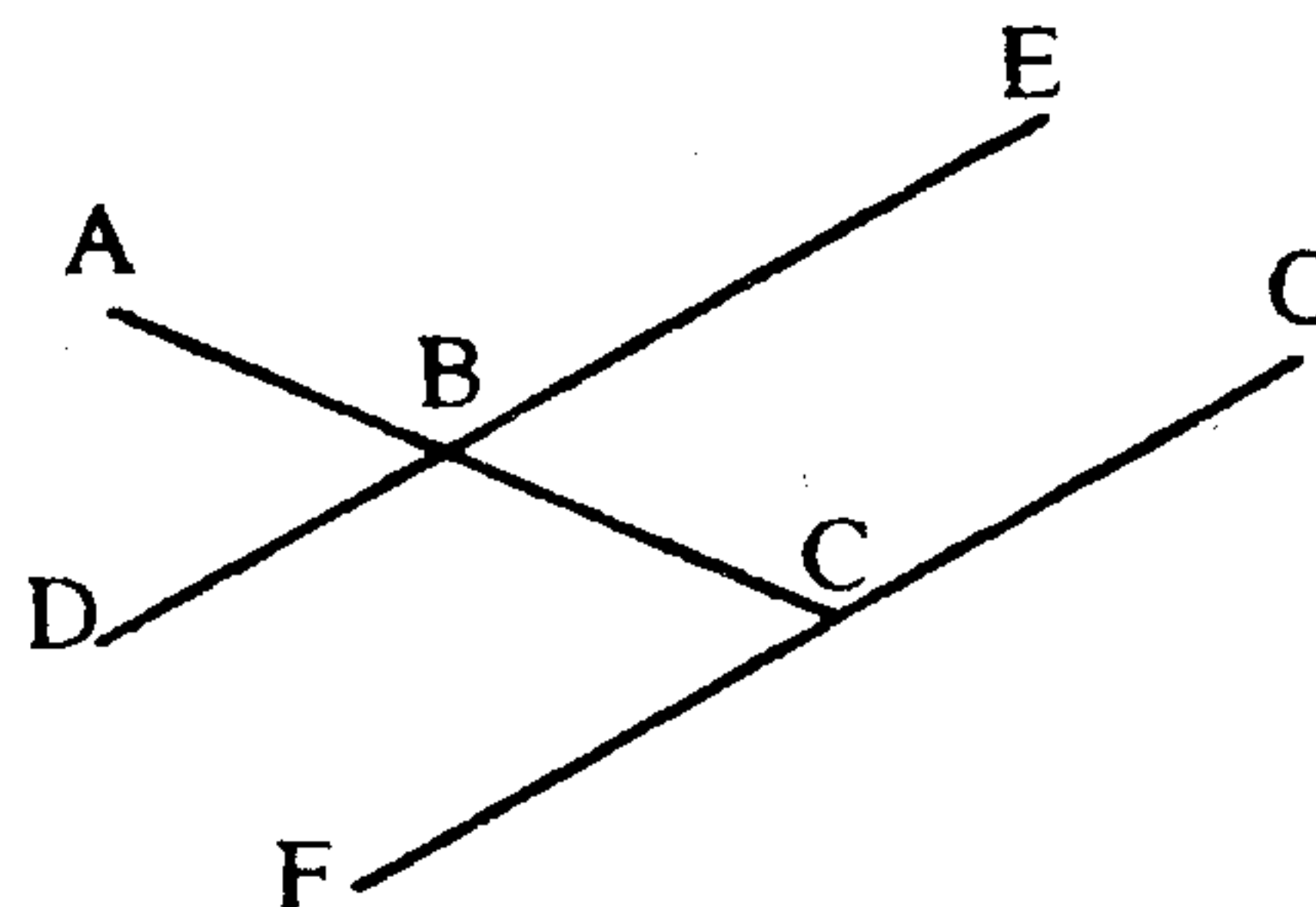
- a) Differentiate: (i) $x^2 e^x$ (ii) $\ln(\cos x)$

- b) The shaded figure, PQR is a sector of a circle, radius 2 cm.
Arc PR is 3 cm in length.



- Find angle $\hat{PQR}$ in radians.
- Find the area of the sector.

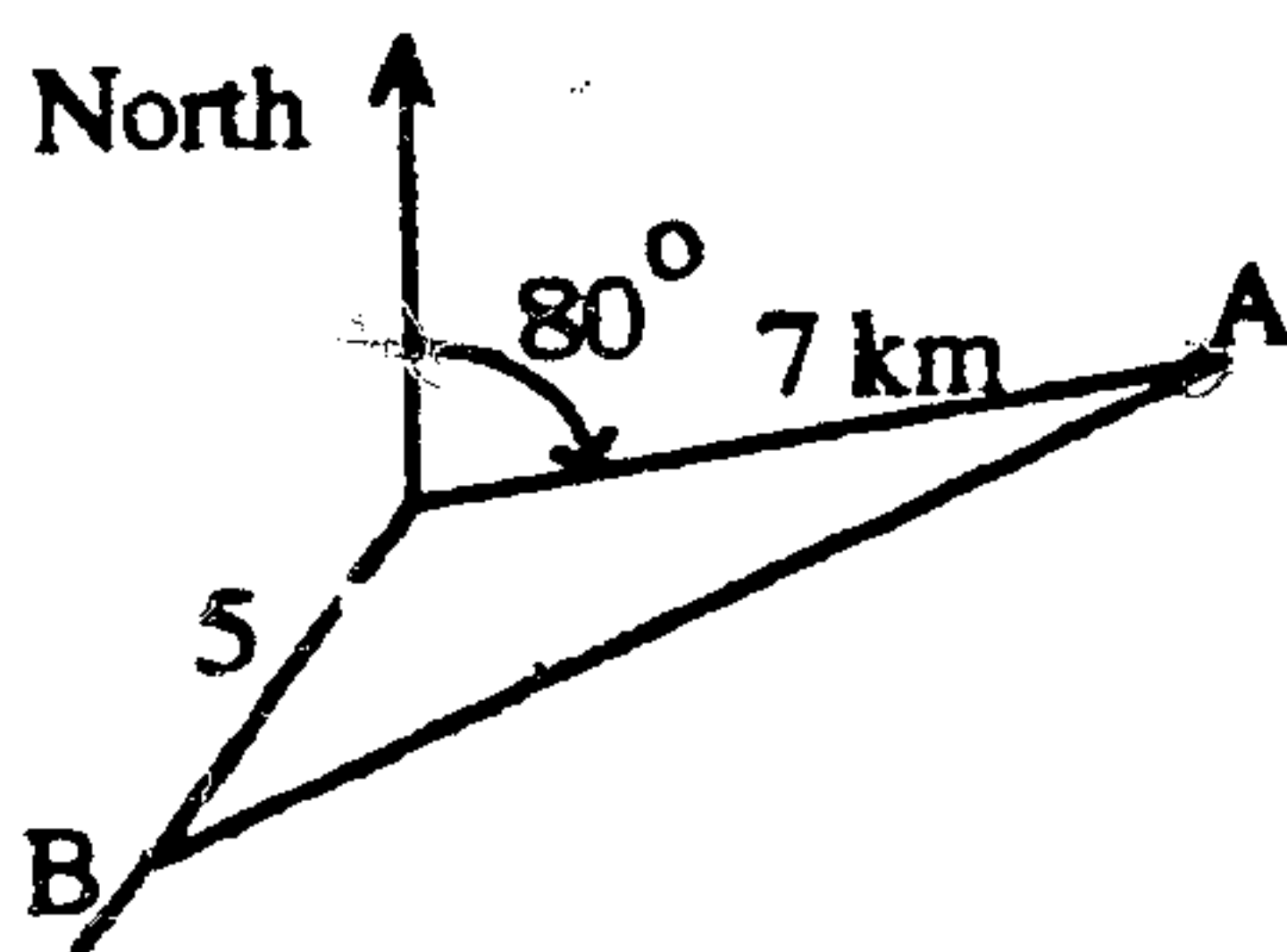
- c) $\angle ABD = 47^\circ$ and $\angle BCG = 133^\circ$
Prove that lines DE and FG are parallel.



- d) Simplify (i) $\sin(\pi - x)$ (ii) $\ln e^{3x}$

Question 3. (Start a new page.)

- a) (i) Integrate $(4x+3)^5$.
 (ii) Find the exact value of $\int_0^{\pi/6} \cos 2x \, dx$.
- b) A ship A is 7 km away from a lighthouse L on a bearing of 080° and a ship B is 5 km away from the lighthouse on a bearing 210° as shown in the diagram.

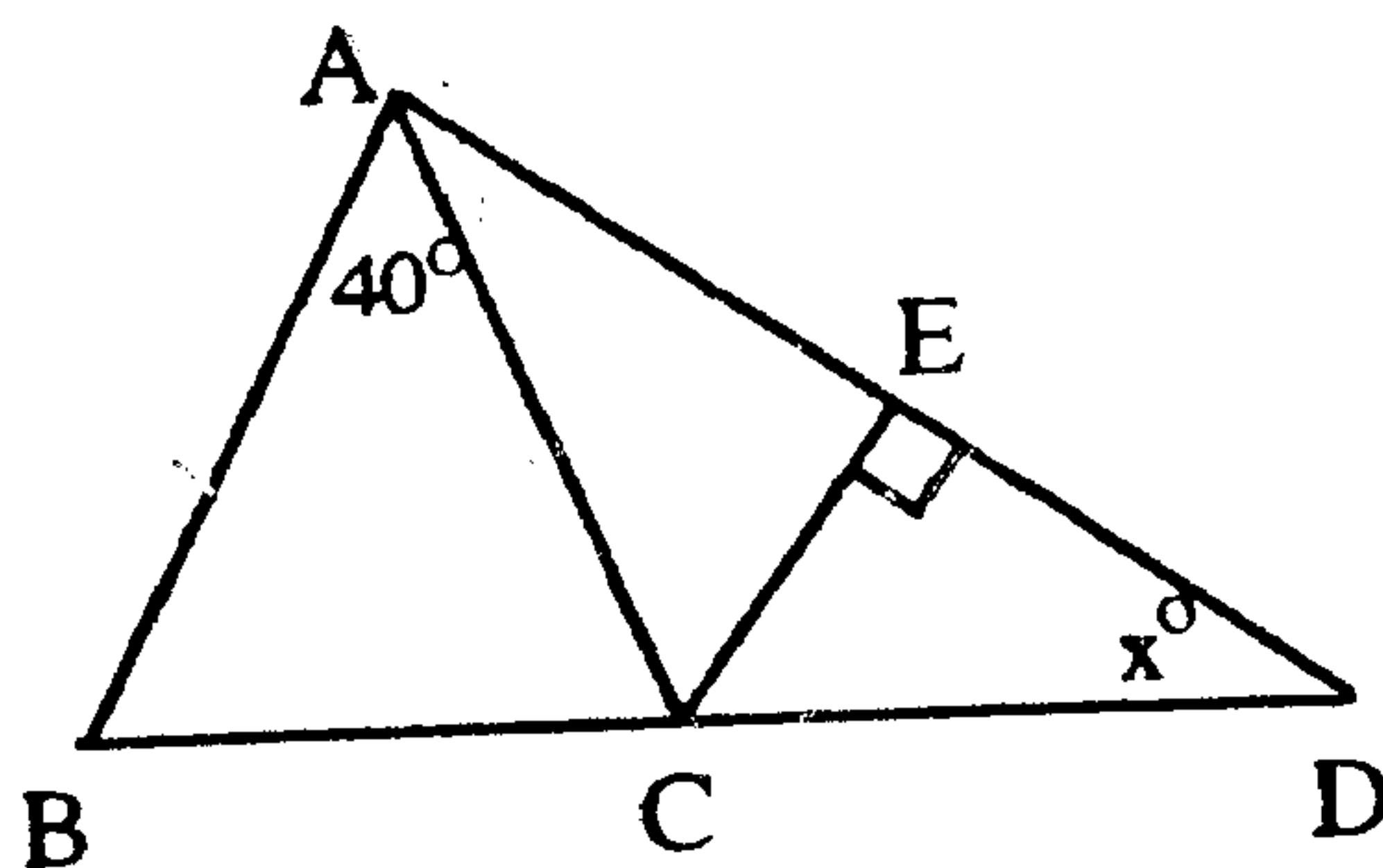


- (i) Find the distance between the ships A and B to the nearest km.
 (ii) Find the bearing of the ship A from the ship B, to the nearest degree.
- c) Solve $2\cos 2x = 1$ for $0 \leq x \leq 180$.

Question 4 (Start a new page.)

a)

(not to scale)



$AB = AC$
 CE bisects $\angle ACD$
 $CE \perp AD$.

- (i) Find x° , giving reasons.
 (ii) Prove that triangle ACD is isosceles.
- b) The gradient of a curve is given by $\frac{dy}{dx} = 3x^2 - 4$.
- (i)(a) Find $\frac{d^2y}{dx^2}$
 (b) Find the values of x for which the curve both increases and is concave downwards. Give reasons for your answer.
- (ii) If the curve passes through the point $(1, -2)$, find the equation of the curve.

Question 5 (Start a new page.)

- a) Consider the curve $y = 4x^3 - 3x^4$.
- (i) Find the points where the curve cuts the x -axis.
 - (ii) Find any stationary points and determine their nature.
 - (iii) Find all points of inflexion.
 - (iv) Sketch the curve showing the above results.
- b) Find the area defined by the curve $y = 4x^3 - 3x^4$ and the x -axis.

Question 6. (Start a new page.)

- a) To start a game a player has to throw a 6 with a die.
Find the probability that the player starts at :
- (i) his first throw.
 - (ii) his second throw.
- b) The displacement of the piston shown, starting from the middle of the cylinder, is modelled by the equation $x(t) = \frac{1}{4}\sin(\pi t)$,
where x is in metres and t in seconds.



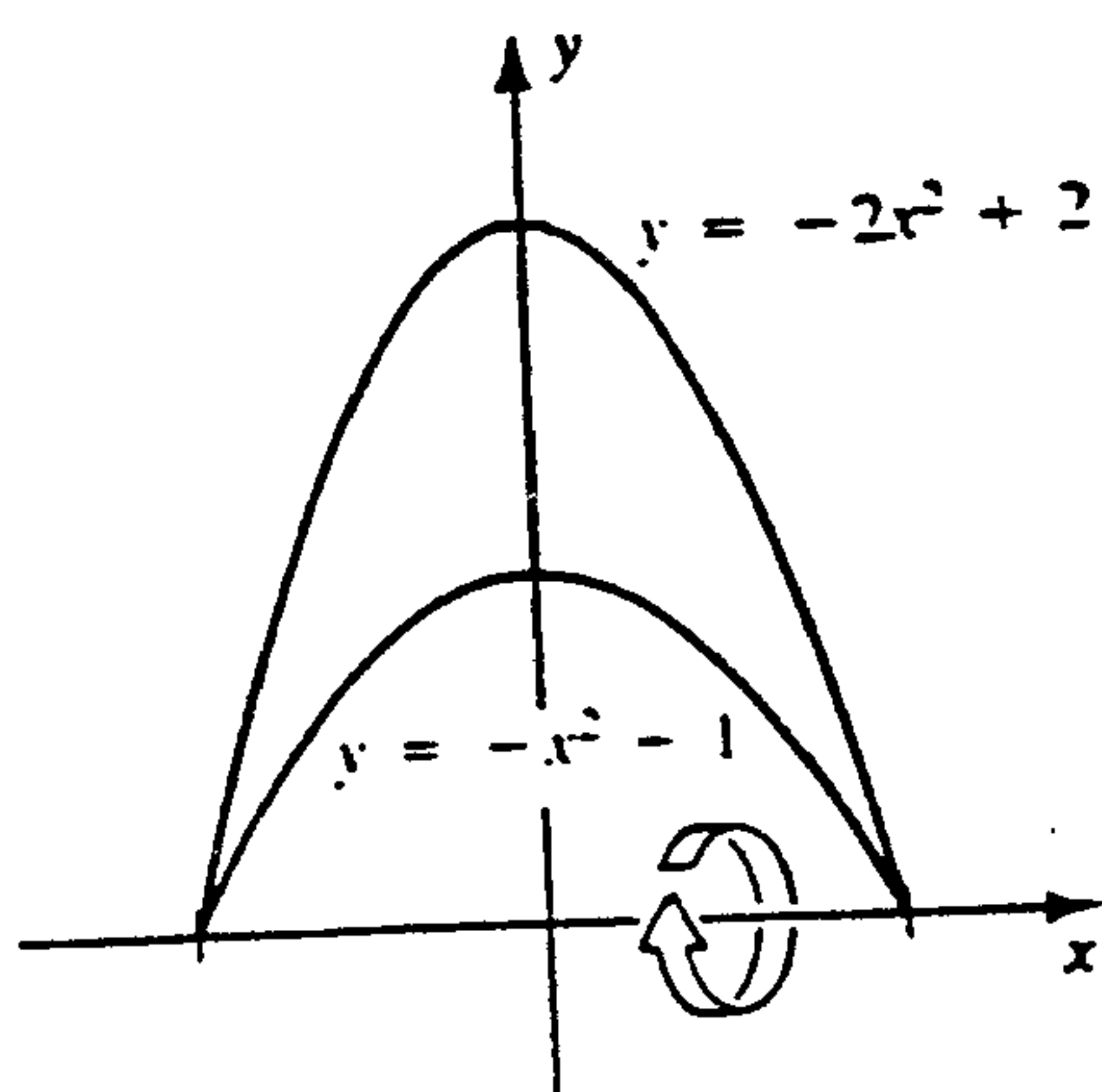
- (i) Sketch the displacement-time function for the first 2 seconds.
 - (ii) Calculate the total distance over which the piston moves in the first 5 seconds.
 - (iii) Calculate the maximum speed of at which the piston moves.
- c) A tree was 12 metres high at the start of a year and it increased by 1.5 metres during that year. If in each succeeding year the growth is $\frac{3}{4}$ of that during the previous year, find the limiting height.

Question 7 (Start a new page.)

a) A body moves in a straight line. At time t seconds its acceleration is given by $a = 6t + 1$. At $t = 0$ the body is at the origin and its velocity is -2 m/s.

- (i) Show that the velocity is given by $v = 3t^2 + t - 2$.
- (ii) Determine when the particle is at rest.
- (iii) Describe the motion.

b)



The area between the two parabolas $y = -2x^2 + 2$ and $y = -x^2 + 1$ is rotated about the x axis.

Find the volume of the solid thus generated.

Question 8 (Start a new page.)

- a)
 - (i) Simplify $\log_2 18 - 2 \log_2 \sqrt{3}$.
 - (ii) Sketch the graph of $y = \ln x$.
 - (iii) Solve the equation $2 \ln x - \ln(2x + 3) = 0$.

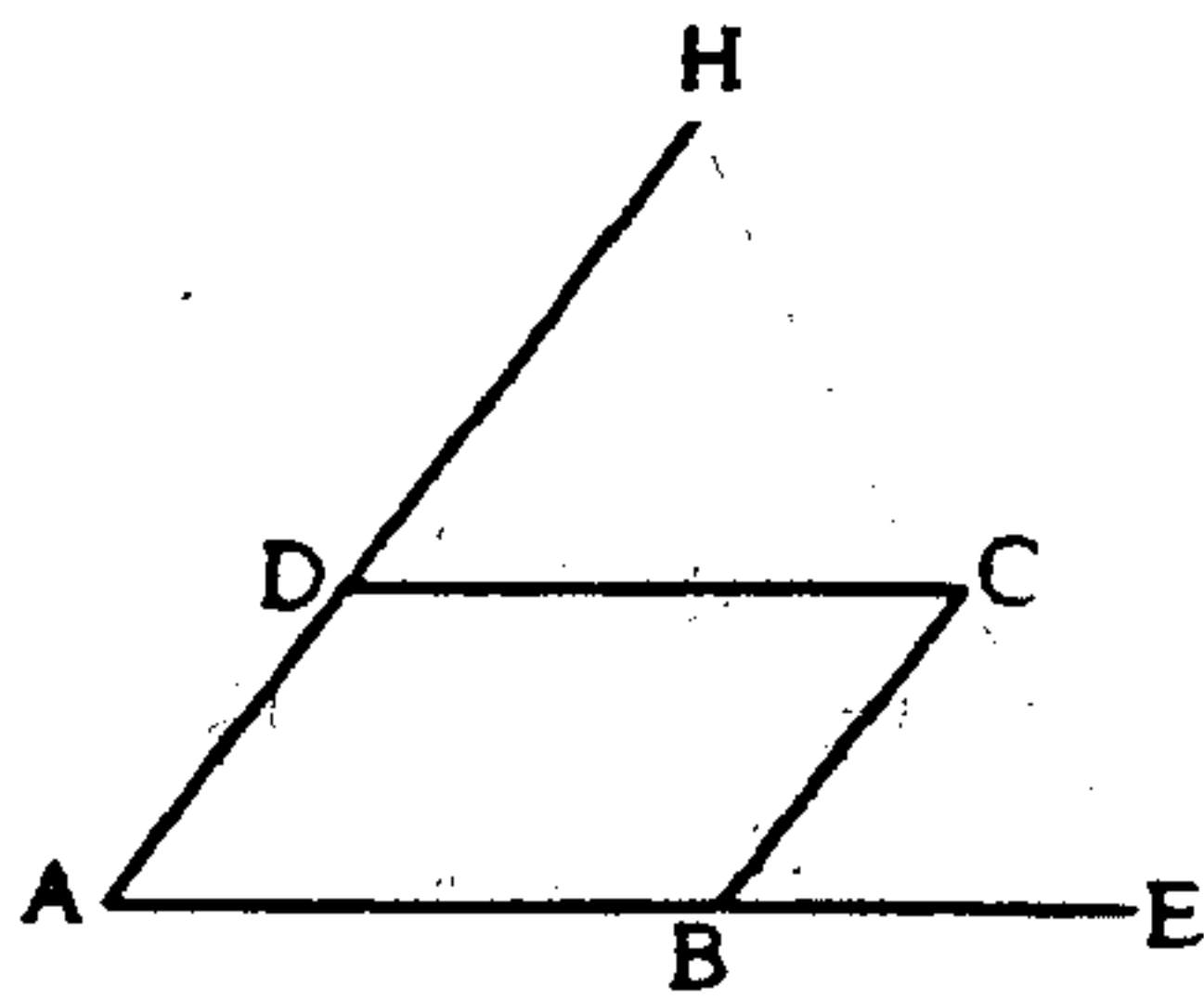
b) A school is divided in two parts : Senior school with 400 boys and 200 girls and Junior school with 400 girls and 300 boys. A first student is chosen at random from the whole school. If this student comes from the Junior school, a second student is chosen from the Senior school; if the first student comes from the Senior school then a second student is chosen from the Junior school. By making use of a tree diagram, or otherwise, find the probability that :

- (i) the second student chosen will be a girl;
- (ii) if the second student chosen is a boy, he is from the senior school.

Question 9

(Start a new page.)

- a) In a parallelogram ABCD the sides AB and AD are extended to E and H respectively so that $\frac{AD}{DH} = \frac{BE}{AB}$. Prove that :



- (i) the triangles HDC and CBE are similar;
(ii) the points H, C and E lie on a straight line.

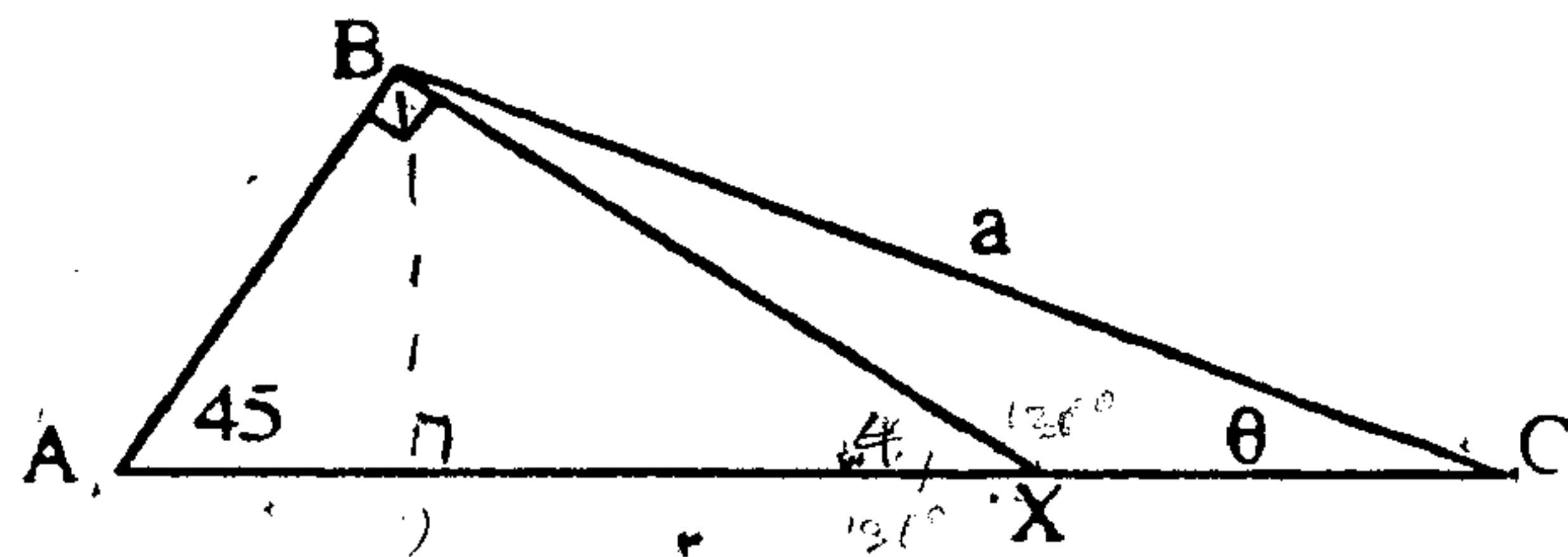
- b) A prize fund is set up with a single investment of \$20,000, to provide an annual prize of \$1,500. The fund accrues interest at 5% per annum, paid yearly. If the first prize is awarded one year after the investment, find :

- (i) the value of the prize fund immediately after the first prize has been awarded;
(ii) the number of years for which the full prize can be awarded.

Question 10

(Start a new page.)

- a) (i) In $\triangle ABC$, BX is perpendicular to AC. Prove that $XC = a(\cos \theta - \sin \theta)$.
(ii) Hence find the exact value of XC when $a = 4$ cm and $\theta = 30^\circ$.



(not to scale)

- b) One thousand trout, each one a year old, are introduced into a large pond. The number still alive after t years is predicted to be

$$N = 1000 e^{-0.205t}$$

- (i) Show that the number of trout decreases at a rate proportional to the number of trout alive.
(ii) The weight $W(t)$ (in kg) of an individual trout is expected to increase according to the formula $W(t) = 0.1 + 0.5t$.
After approximately how many years is the total weight (in kg) of all the trout in the pond a maximum ?

The End

Q1: (i) $m \times m = -1$
 (a) $-3 \times \frac{1}{3} = -1 \therefore PQ \perp QR$

(ii) $PQ = \sqrt{10}$
 $QR = \sqrt{10} \therefore \triangle PQR$ is isosceles

(iii) $S(4,4)$

(b) $x = \frac{8}{3}$

(c) $A = 9\sqrt{2}$

(d) \$11,658

Q2:

(a) (i) $xe^{2x}(2+x)$

(ii) $-\tan x$

(b) (i) 1.5°

(ii) 3cm^2

(c) —

(d) (i) $\sin x$

(ii) $3x$

Q3: (i) $\frac{(4x+3)^6}{24} + C$

(a) (ii) $\frac{\sqrt{3}}{4}$

(b) (i) 11kms

(ii) 59°

(c) $x = 30^\circ, 150^\circ$

Q4: (i) $x = 35^\circ$

(a) (ii) —

(b) (i) bx

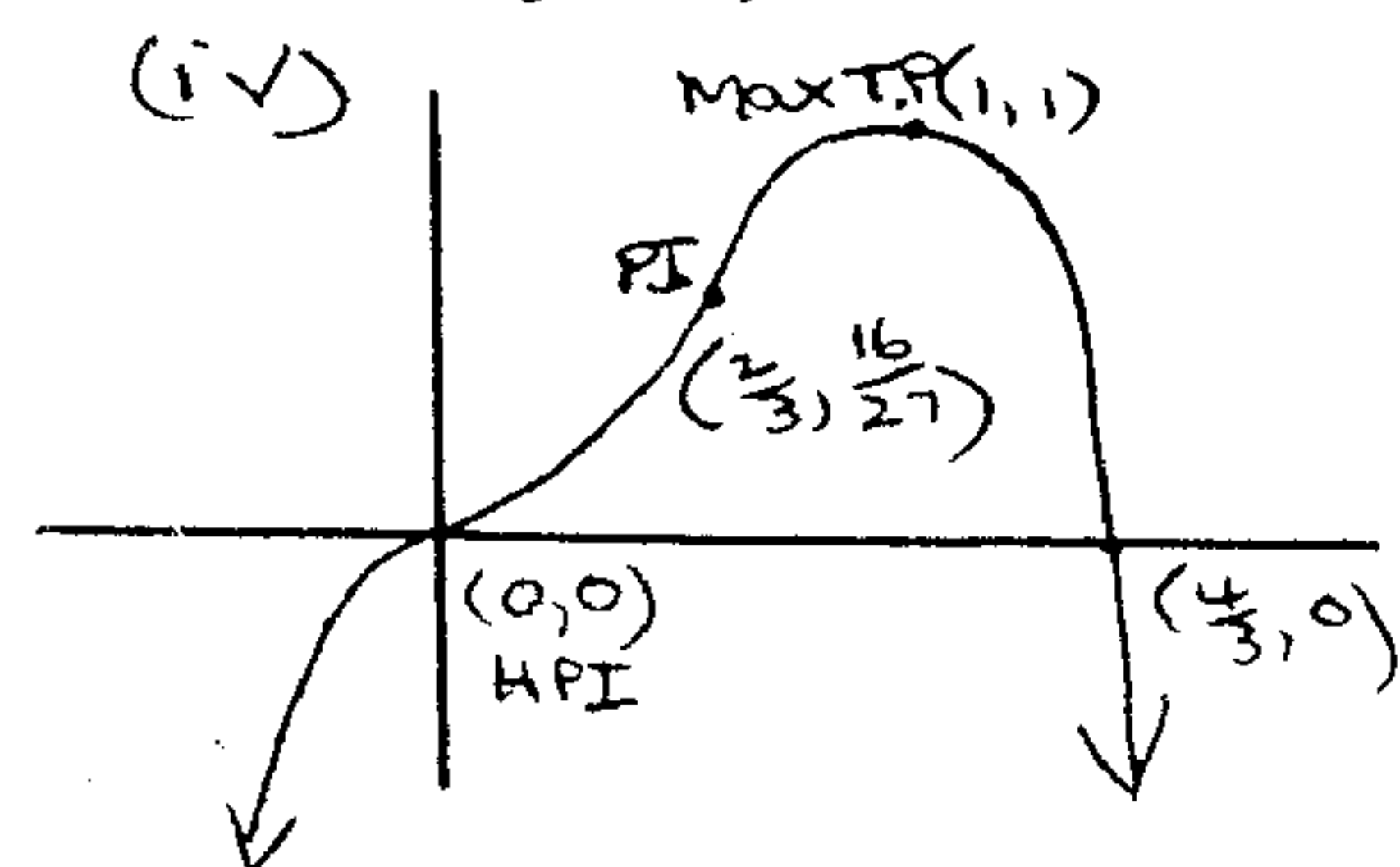
(ii) $x < -\frac{2\sqrt{3}}{3}$

(iii) $y = x^3 - 4x + 1$

Q5: (i) $(0,0), (\frac{4}{3}, 0)$

(a) (ii) $(0,0)$ Horizontal P.I.
 $(1,1)$ Max T.P.

(iii) $(0,0)$ already found
 $(\frac{2}{3}, \frac{16}{27})$

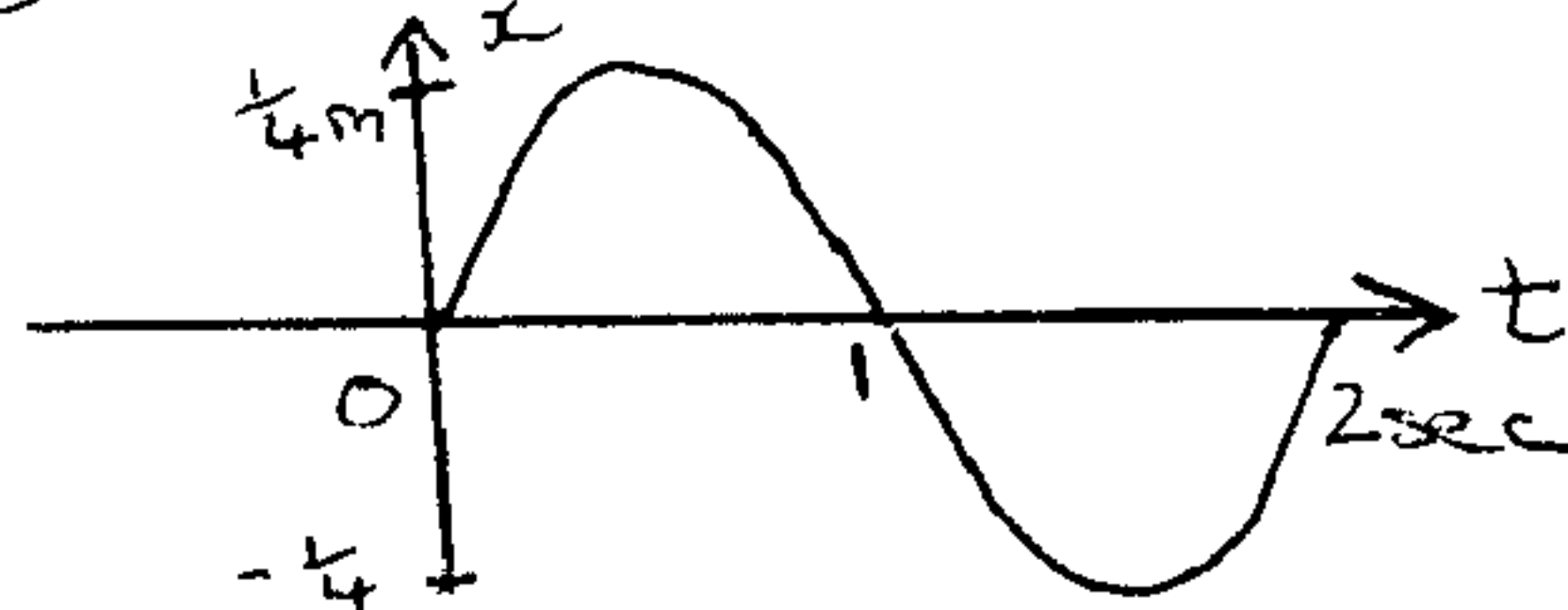


(b) $\frac{256}{405} v^2$

Q6: (i) $\frac{1}{6}$

(a) (ii) $\frac{5}{36}$

(b) (i) $x = \frac{1}{4} \sin(\pi t)$



(ii) 2km

(iii) $\frac{\pi}{4} \text{ m/s}$

(c) 18m

Q7: (i) —

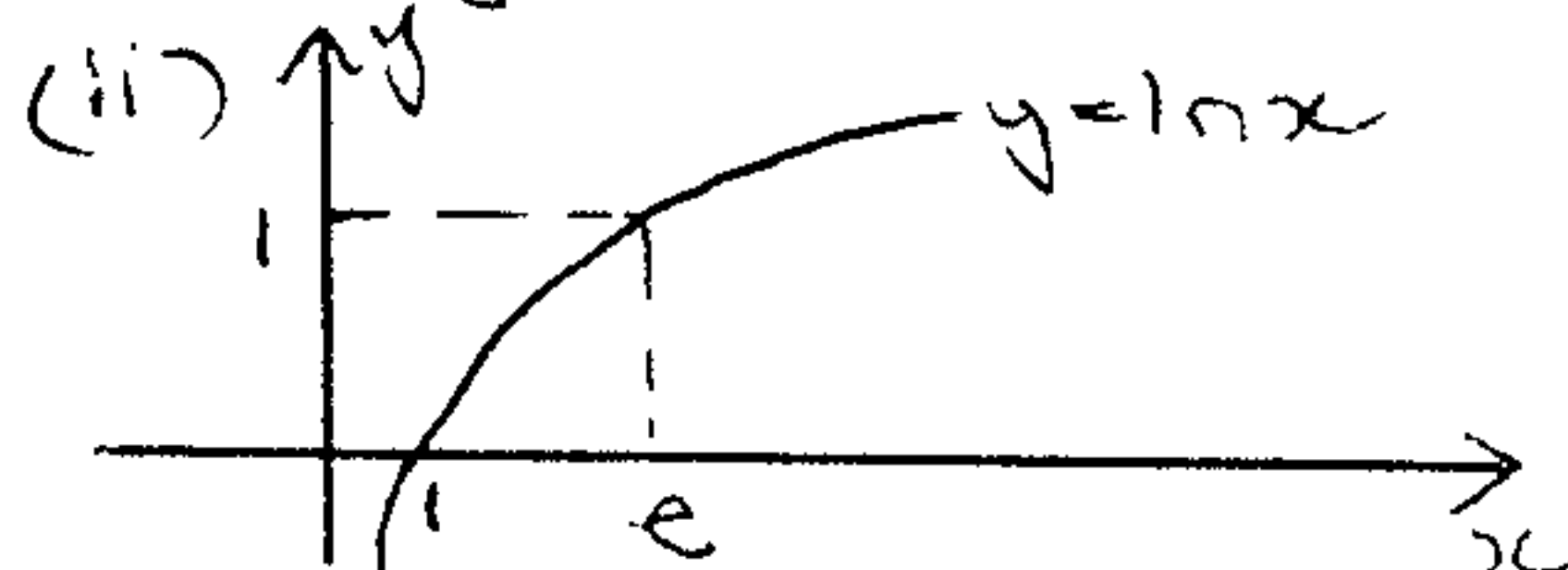
(a) (ii) $t = \frac{2}{3} \text{ sec}$

(iii) Initially particle is at the origin and moves to the left, slowing down (since force to the right opposing motion). Particle stops at $t = \frac{2}{3} \text{ sec}$ and then changes direction moving now towards the right.

For (b) $V = \frac{16\pi}{5} v^3$

Q8:

(a) (i) $\log_2 6$



(iii) $x = 3$

(b) (i) $\frac{121}{273}$

(ii) $\frac{49}{76}$

Q9:

(a) (i) —

(ii) —

(b) (i) \$19,500

(ii) 22 yrs

Q10:

(a) (i) —

(ii) $2\sqrt{3} - 2$

(b) (i) $\frac{dN}{dt} = -0.205N$

(ii) 4.7 yrs.